

30-Day Placement DSA Plan

Day 1 — Basics + Recursion

#	Problem	Striver Topic	Striver Subtopic	☐
1	Print 1 to N	Learn Basic Recursion	Learn Basic Recursion	☐
2	Print N to 1	Learn Basic Recursion	Learn Basic Recursion	☐
3	Sum of First N Numbers	Learn Basic Recursion	Learn Basic Recursion	☐
4	Factorial	Learn Basic Recursion	Learn Basic Recursion	☐
5	Reverse an Array using Recursion	Learn Basic Recursion	Learn Basic Recursion	☐
6	Check Palindrome	Learn Basic Recursion	Learn Basic Recursion	☐

Day 2 — Arrays I

#	Problem	Striver Topic	Striver Subtopic	☐
7	Largest Element in an Array	Solve Problems on Arrays [Easy -> Medium -> Hard]	Easy	☐
8	Second Largest Element	Solve Problems on Arrays [Easy -> Medium -> Hard]	Easy	☐
9	Check if Array is Sorted	Solve Problems on Arrays [Easy -> Medium -> Hard]	Easy	☐
10	Remove Duplicates from Sorted Array	Solve Problems on Arrays [Easy -> Medium -> Hard]	Easy	☐
11	Left Rotate Array by One	Solve Problems on Arrays [Easy -> Medium -> Hard]	Easy	☐
12	Left Rotate Array by K Places	Solve Problems on Arrays [Easy -> Medium -> Hard]	Easy	☐

Day 3 — Arrays II + Hashing

#	Problem	Striver Topic	Striver Subtopic	☐
13	Two Sum	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
14	Majority Element	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
15	Find Missing Number	Solve Problems on Arrays [Easy -> Medium -> Hard]	Easy	☐
16	Number Appearing Once	Solve Problems on Arrays [Easy -> Medium -> Hard]	Easy	☐
17	Longest Consecutive Sequence	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
18	Count Frequencies of Array Elements	Learn Basic Hashing	Learn Basic Hashing	☐

Day 4 — Prefix Sum + Kadane

#	Problem	Striver Topic	Striver Subtopic	☐
19	Maximum Subarray — Kadane	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
20	Best Time to Buy and Sell Stock	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
21	Longest Subarray With Sum K	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
22	Subarray Sum Equals K	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
23	Rearrange Array Elements by Sign	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐

Day 5 — Arrays III

#	Problem	Striver Topic	Striver Subtopic	☐
24	Sort Colors	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
25	3Sum	Solve Problems on Arrays [Easy -> Medium -> Hard]	Hard	☐
26	4Sum	Solve Problems on Arrays [Easy -> Medium -> Hard]	Hard	☐
27	Maximum Subarray — Revision	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
28	Set Matrix Zeroes	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐
29	Rotate Image	Solve Problems on Arrays [Easy -> Medium -> Hard]	Medium	☐

Day 6 — Two Pointers

#	Problem	Striver Topic	Striver Subtopic	☐
30	Two Sum II	[NOT IN A2Z] LeetCode	Arrays / Two Pointers	☐
31	Remove Duplicates	[NOT IN A2Z] LeetCode	Arrays / Two Pointers	☐
32	Container With Most Water	[NOT IN A2Z] LeetCode	Arrays / Two Pointers	☐
33	Trapping Rain Water	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Monotonic Stack/Queue Problems [VVV. Imp]	☐

Day 7 — Sliding Window

#	Problem	Striver Topic	Striver Subtopic	☐
34	Maximum Sum Subarray of Size K	[NOT IN A2Z] GeeksforGeeks	Sliding Window	☐
35	Longest Substring Without Repeating Characters	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Sliding Window & Two Pointer Combined Problems	☐
36	Longest Repeating Character Replacement	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Sliding Window & Two Pointer Combined Problems	☐
37	Minimum Window Substring	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Sliding Window & Two Pointer Combined Problems	☐

Day 8 — Binary Search I

#	Problem	Striver Topic	Striver Subtopic	☐
38	Binary Search	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐
39	Lower Bound	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐
40	Upper Bound	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐
41	Search Insert Position	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐
42	First and Last Position	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐
43	Count Occurrences in Sorted Array	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐

Day 9 — Binary Search II

#	Problem	Striver Topic	Striver Subtopic	☐
44	Search in Rotated Sorted Array	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐
45	Find Minimum in Rotated Sorted Array	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐
46	Single Element in Sorted Array	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐
47	Find Peak Element	Binary Search [1D, 2D Arrays, Search Space]	BS on 1D Arrays	☐
48	Koko Eating Bananas	Binary Search [1D, 2D Arrays, Search Space]	BS on Answers	☐
49	Capacity to Ship Packages	Binary Search [1D, 2D Arrays, Search Space]	BS on Answers	☐

Day 10 — Sorting

#	Problem	Striver Topic	Striver Subtopic	☐
50	Selection Sort	Learn Important Sorting Techniques	Sorting-I	☐
51	Bubble Sort	Learn Important Sorting Techniques	Sorting-I	☐
52	Insertion Sort	Learn Important Sorting Techniques	Sorting-I	☐
53	Merge Sort	Learn Important Sorting Techniques	Sorting-II	☐
54	Quick Sort	Learn Important Sorting Techniques	Sorting-II	☐
55	Sort an Array	[NOT IN A2Z] LeetCode	Sorting	☐

Day 11 — Linked List I

#	Problem	Striver Topic	Striver Subtopic	☐
56	Implement Singly Linked List	[NOT IN A2Z] GeeksforGeeks	Learn 1D LinkedList	☐
57	Reverse Linked List	Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]	Medium Problems of LL	☐
58	Middle of Linked List	Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]	Medium Problems of LL	☐
59	Detect Cycle	Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]	Medium Problems of LL	☐
60	Length of Loop	Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]	Medium Problems of LL	☐

#	Problem	Striver Topic	Striver Subtopic	☐
61	Palindrome Linked List	Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]	Medium Problems of LL	☐

Day 12 — Linked List II

#	Problem	Striver Topic	Striver Subtopic	☐
62	Remove Nth Node From End	Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]	Medium Problems of LL	☐
63	Intersection of Two Linked Lists	Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]	Medium Problems of LL	☐
64	Merge Two Sorted Lists	[NOT IN A2Z] LeetCode	Linked List	☐
65	Add Two Numbers	Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]	Medium Problems of LL	☐
66	Rotate Linked List	Learn LinkedList [Single LL, Double LL, Medium, Hard Problems]	Hard Problems of LL	☐

Day 13 — Stack + Queue

#	Problem	Striver Topic	Striver Subtopic	☐
67	Implement Stack	[NOT IN A2Z] GeeksforGeeks	Learning	☐
68	Implement Queue	[NOT IN A2Z] GeeksforGeeks	Learning	☐
69	Valid Parentheses	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Learning	☐
70	Min Stack	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Learning	☐
71	Queue Using Stacks	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Learning	☐
72	Stack Using Queues	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Learning	☐

Day 14 — Monotonic Stack

#	Problem	Striver Topic	Striver Subtopic	☐
73	Next Greater Element	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Monotonic Stack/Queue Problems [VVV. Imp]	☐
74	Next Greater Element II	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Monotonic Stack/Queue Problems [VVV. Imp]	☐
75	Daily Temperatures	[NOT IN A2Z] LeetCode	Monotonic Stack	☐
76	Largest Rectangle in Histogram	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Monotonic Stack/Queue Problems [VVV. Imp]	☐
77	Maximal Rectangle	Stack and Queues [Learning, Pre-In-Post-fix, Monotonic Stack, Implementation]	Monotonic Stack/Queue Problems [VVV. Imp]	☐

Day 15 — Backtracking I

#	Problem	Striver Topic	Striver Subtopic	☐
78	Subsets	Recursion [PatternWise]	Subsequences Pattern	☐
79	Subsets II	Recursion [PatternWise]	Subsequences Pattern	☐
80	Permutations	[NOT IN A2Z] LeetCode	Backtracking	☐
81	Combination Sum	Recursion [PatternWise]	Subsequences Pattern	☐

Day 16 — Backtracking II

#	Problem	Striver Topic	Striver Subtopic	☐
82	Combination Sum II	Recursion [PatternWise]	Subsequences Pattern	☐
83	Letter Combinations of Phone Number	Recursion [PatternWise]	Subsequences Pattern	☐
84	Palindrome Partitioning	Recursion [PatternWise]	Trying out all Combos / Hard	☐
85	N-Queens	Recursion [PatternWise]	Trying out all Combos / Hard	☐

Day 17 — Binary Tree Basics

#	Problem	Striver Topic	Striver Subtopic	☐
86	Preorder Traversal	Binary Trees [Traversals, Medium and Hard Problems]	Traversals	☐
87	Inorder Traversal	Binary Trees [Traversals, Medium and Hard Problems]	Traversals	☐
88	Postorder Traversal	Binary Trees [Traversals, Medium and Hard Problems]	Traversals	☐
89	Level Order Traversal	Binary Trees [Traversals, Medium and Hard Problems]	Traversals	☐
90	Maximum Depth of Binary Tree	Binary Trees [Traversals, Medium and Hard Problems]	Medium Problems	☐

Day 18 — Binary Tree Patterns

#	Problem	Striver Topic	Striver Subtopic	☐
91	Balanced Binary Tree	Binary Trees [Traversals, Medium and Hard Problems]	Medium Problems	☐
92	Diameter of Binary Tree	Binary Trees [Traversals, Medium and Hard Problems]	Medium Problems	☐
93	Maximum Path Sum	Binary Trees [Traversals, Medium and Hard Problems]	Medium Problems	☐
94	Same Tree	Binary Trees [Traversals, Medium and Hard Problems]	Medium Problems	☐
95	Symmetric Tree	Binary Trees [Traversals, Medium and Hard Problems]	Medium Problems	☐

Day 19 — Tree Advanced + BST

#	Problem	Striver Topic	Striver Subtopic	☐
96	Lowest Common Ancestor	Binary Trees [Traversals, Medium and Hard Problems]	Hard Problems	☐
97	Binary Tree Right Side View	Binary Trees [Traversals, Medium and Hard Problems]	Medium Problems	☐
98	Validate BST	Binary Search Trees [Concept and Problems]	Practice Problems	☐
99	Search in BST	Binary Search Trees [Concept and Problems]	Concepts	☐
100	Kth Smallest Element in BST	Binary Search Trees [Concept and Problems]	Practice Problems	☐

Day 20 — Heap

#	Problem	Striver Topic	Striver Subtopic	☐
101	Kth Largest Element	Heaps [Learning, Medium, Hard Problems]	Medium Problems	☐
102	Top K Frequent Elements	Heaps [Learning, Medium, Hard Problems]	Hard Problems	☐
103	K Closest Points to Origin	[NOT IN A2Z] LeetCode	Heap / Priority Queue	☐
104	Merge K Sorted Lists	Heaps [Learning, Medium, Hard Problems]	Medium Problems	☐
105	Find Median From Data Stream	Heaps [Learning, Medium, Hard Problems]	Hard Problems	☐

Day 21 — Greedy + Intervals

#	Problem	Striver Topic	Striver Subtopic	☐
106	Assign Cookies	Greedy Algorithms [Easy, Medium/Hard]	Easy Problems	☐
107	Jump Game	Greedy Algorithms [Easy, Medium/Hard]	Medium/Hard	☐
108	Jump Game II	Greedy Algorithms [Easy, Medium/Hard]	Medium/Hard	☐
109	Merge Intervals	Greedy Algorithms [Easy, Medium/Hard]	Medium/Hard	☐
110	Non-overlapping Intervals	Greedy Algorithms [Easy, Medium/Hard]	Medium/Hard	☐
111	Meeting Rooms II	[NOT IN A2Z] LeetCode	Greedy / Intervals	☐

Day 22 — Graph BFS/DFS

#	Problem	Striver Topic	Striver Subtopic	☐
112	Number of Islands	Graphs [Concepts & Problems]	Problems on BFS/DFS	☐
113	Flood Fill	Graphs [Concepts & Problems]	Problems on BFS/DFS	☐
114	Number of Provinces	Graphs [Concepts & Problems]	Problems on BFS/DFS	☐
115	Clone Graph	[NOT IN A2Z] LeetCode	Graph BFS/DFS	☐

Day 23 — Graph Patterns

#	Problem	Striver Topic	Striver Subtopic	☐
116	Rotting Oranges	Graphs [Concepts & Problems]	Problems on BFS/DFS	☐
117	Cycle Detection in Undirected Graph	Graphs [Concepts & Problems]	Problems on BFS/DFS	☐
118	Bipartite Graph	Graphs [Concepts & Problems]	Problems on BFS/DFS	☐
119	Surrounded Regions	Graphs [Concepts & Problems]	Problems on BFS/DFS	☐

Day 24 — Advanced Graph

#	Problem	Striver Topic	Striver Subtopic	☐
120	Course Schedule	Graphs [Concepts & Problems]	Topo Sort and Problems	☐
121	Course Schedule II	Graphs [Concepts & Problems]	Topo Sort and Problems	☐
122	Network Delay Time	Graphs [Concepts & Problems]	Shortest Path Algorithms and Problems	☐
123	Number of Connected Components	Graphs [Concepts & Problems]	Learning	☐

Day 25 — Bit Manipulation + Trie

#	Problem	Striver Topic	Striver Subtopic	☐
124	Single Number	Bit Manipulation [Concepts & Problems]	Interview Problems	☐
125	Number of 1 Bits	Bit Manipulation [Concepts & Problems]	Learn Bit Manipulation	☐
126	Counting Bits	[NOT IN A2Z] LeetCode	Bit Manipulation	☐
127	Power of Two	Bit Manipulation [Concepts & Problems]	Learn Bit Manipulation	☐
128	Implement Trie	Tries	Theory	☐
129	Implement Trie II	Tries	Problems	☐

Day 26 — 1D DP

#	Problem	Striver Topic	Striver Subtopic	☐
130	Climbing Stairs	Dynamic Programming [Patterns and Problems]	1D DP	☐
131	Frog Jump	Dynamic Programming [Patterns and Problems]	1D DP	☐
132	House Robber	Dynamic Programming [Patterns and Problems]	1D DP	☐
133	House Robber II	[NOT IN A2Z] LeetCode	1D DP	☐

Day 27 — Grid DP

#	Problem	Striver Topic	Striver Subtopic	☐
134	Unique Paths	Dynamic Programming [Patterns and Problems]	2D/3D DP and DP on Grids	☐
135	Unique Paths II	Dynamic Programming [Patterns and Problems]	2D/3D DP and DP on Grids	☐
136	Minimum Path Sum	[NOT IN A2Z] LeetCode	2D/3D DP and DP on Grids	☐
137	Triangle	Dynamic Programming [Patterns and Problems]	2D/3D DP and DP on Grids	☐

Day 28 — Knapsack / Subset

#	Problem	Striver Topic	Striver Subtopic	☐
138	0/1 Knapsack	[NOT IN A2Z] GeeksforGeeks	DP on Subsequences / Knapsack	☐
139	Partition Equal Subset Sum	Dynamic Programming [Patterns and Problems]	DP on Subsequences	☐
140	Target Sum	Dynamic Programming [Patterns and Problems]	DP on Subsequences	☐
141	Coin Change	[NOT IN A2Z] LeetCode	DP on Subsequences / Unbounded Knapsack	☐

Day 29 — DP Subsequences

#	Problem	Striver Topic	Striver Subtopic	☐
142	Longest Common Subsequence	Dynamic Programming [Patterns and Problems]	DP on Strings	☐
143	Longest Increasing Subsequence	Dynamic Programming [Patterns and Problems]	DP on LIS	☐
144	Longest Palindromic Subsequence	Dynamic Programming [Patterns and Problems]	DP on Strings	☐
145	Edit Distance	Dynamic Programming [Patterns and Problems]	DP on Strings	☐

Day 30 — Placement Test

Do not introduce new topics on Day 30. Treat this as a short timed placement checkpoint.

Area	Final Checklist Problem	☐
Array / Hashing	Two Sum or Majority Element	☐
Sliding Window	Longest Substring Without Repeating Characters	☐
Binary Search	Koko Eating Bananas	☐
Linked List	Reverse Linked List	☐
Tree / Graph	Number of Islands	☐
Dynamic Programming	House Robber or LCS	☐

Suggested use: solve each problem without notes first, then review the pattern and mistakes. Tick the box only after completing the problem.